

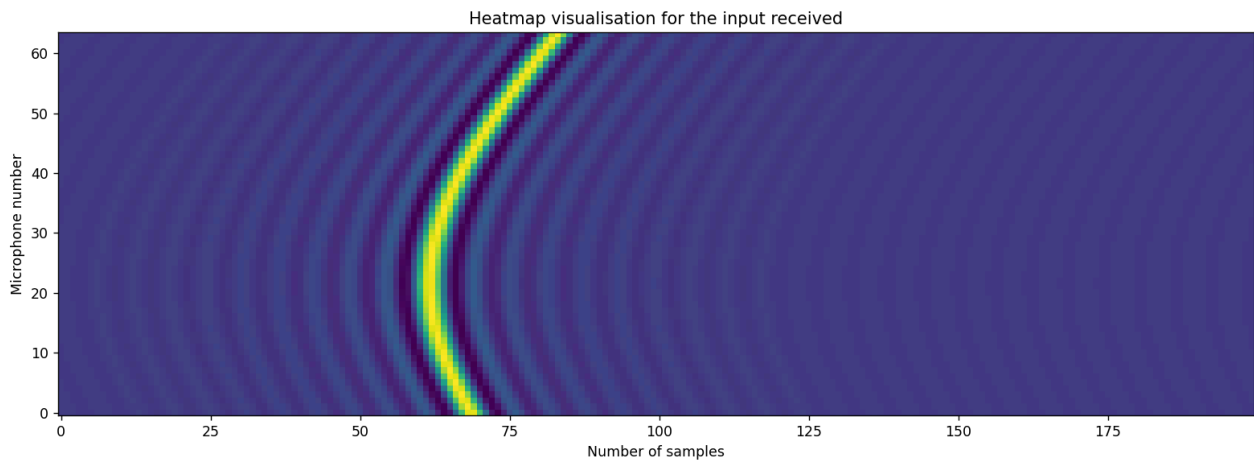
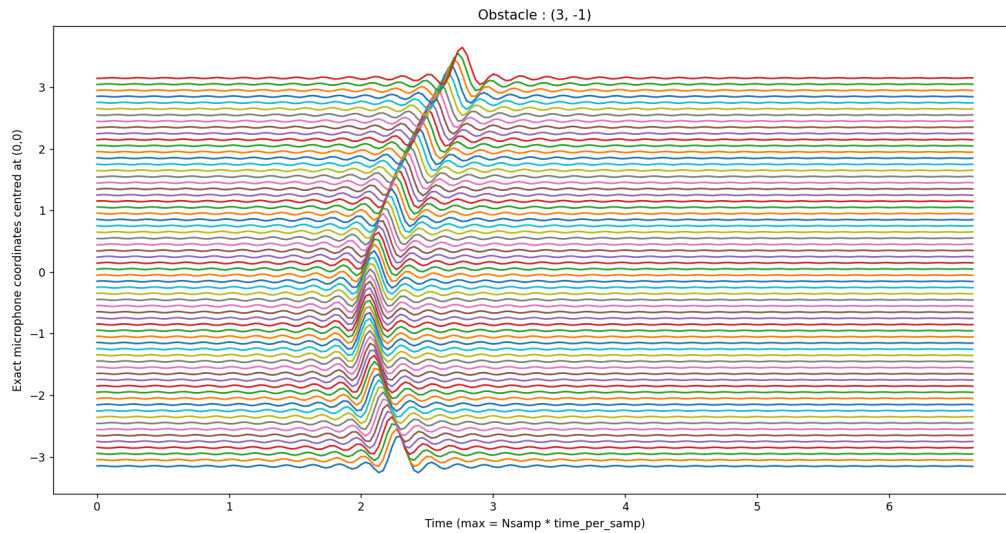
APL Assignment 7 Report : Sound Localization

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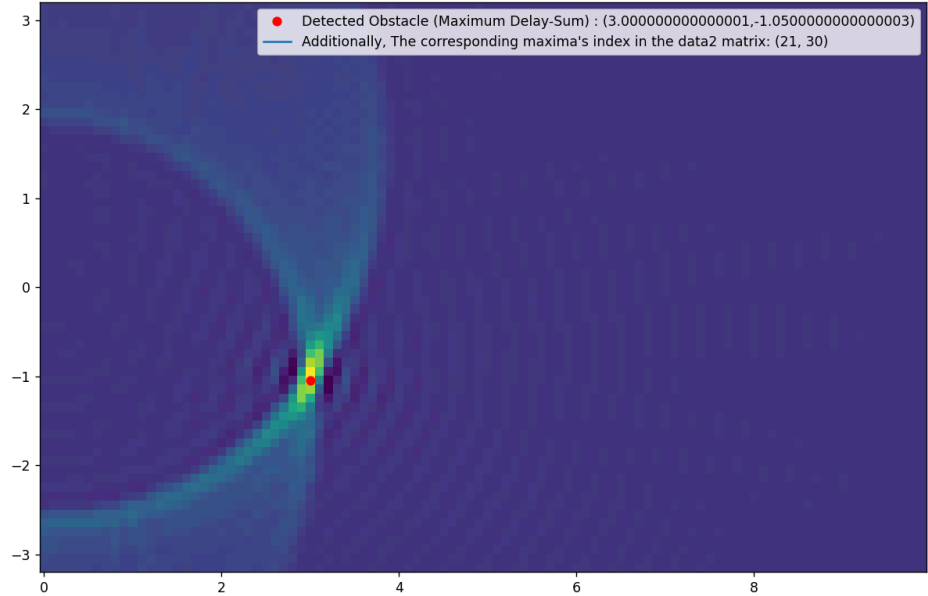
Roll: EE23B097

NOTE: The QUESTIONS have all been answered in this report, and the CODE sections have been done and marked with comments in the source code file.

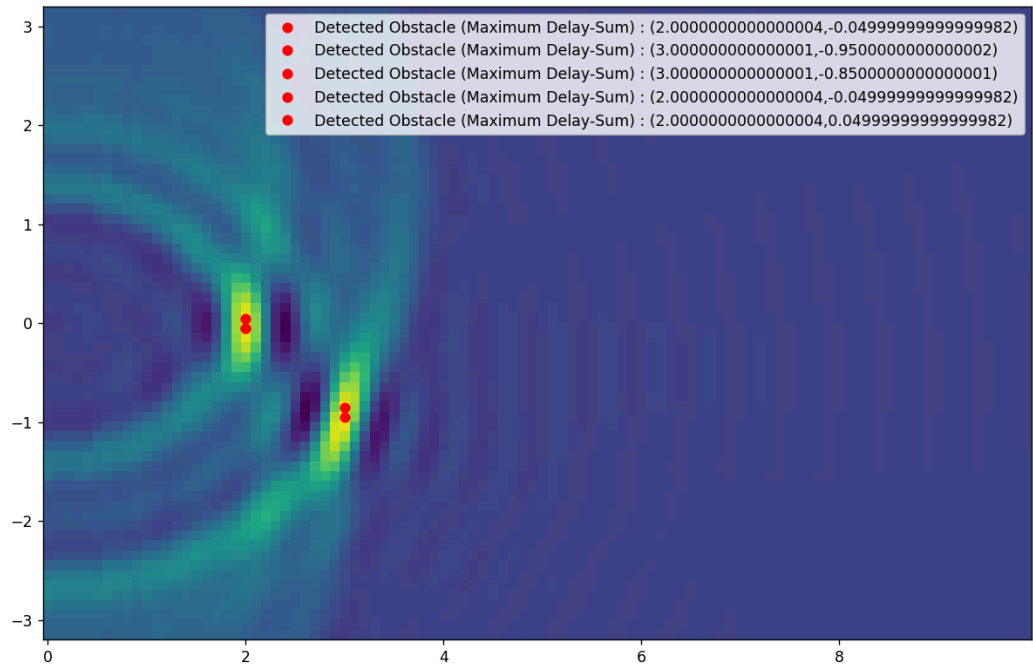
Firstly, the outputs for each step in case of the (3,-1) obstacle case is attached:



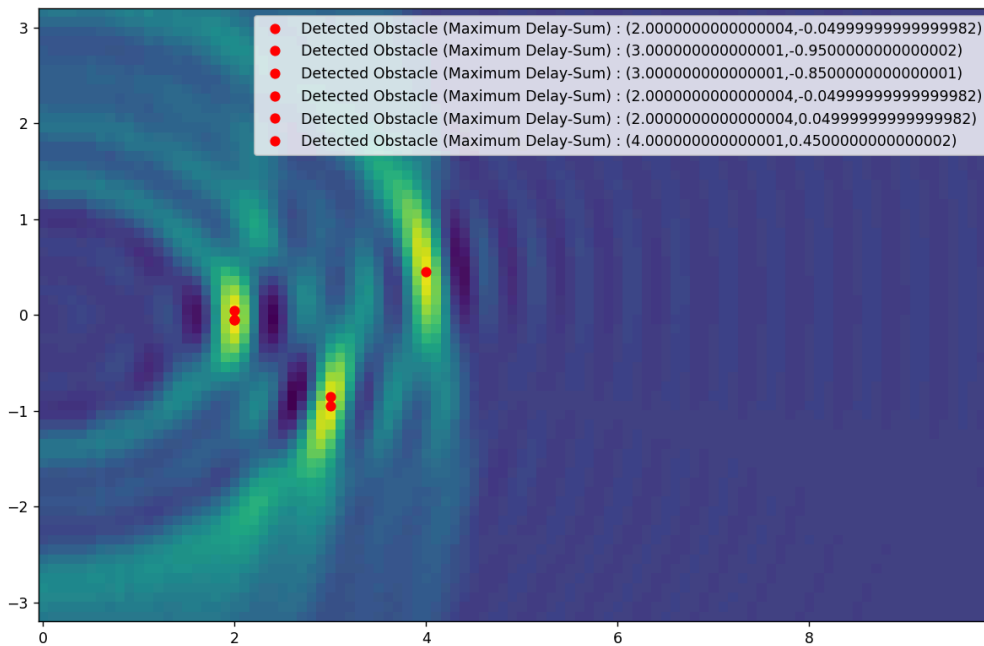
Reconstructed Image (With actual coordinates, not array indices) Nmics: 64, Nsamp: 200, src: (0, 0), pitch: 0.1, dist_per_samp: 0.1, C: 3, SincP: 10, Obstacle: (3, -1)



- The output images for the input files given as **rx2.txt**, **rx3.txt** in the assignment are also given below. Note: The code to find the multiple obstacles is **kinda weird** and prints multiple coordinates as I coded it to mark all points till a certain value of delay-sum below the maximum's:



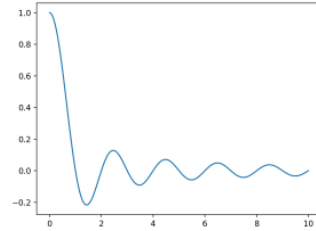
← rx2.txt output



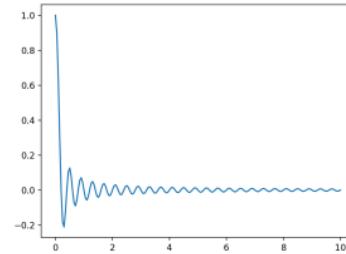
← rx3.txt output

Answers to the QUESTIONS given in assignment description:

- Q1. The plots below show two example Sinc pulses. How will you generate pulses that look like this? Which parameter should be changed? What effect do you think this will have on the final image?
 - A1) These plots can be generated by generating an x array, called t in code. Then using it to make `wsrc(t)` and finally plotting using `plt.plot(t,wsrc(t))`.
 - The SincP parameter in code is what needs to be changed to generate these 2 plots specifically because the maximum x length stays same for both and there is a compression happening. So it is equivalent to converting $f(\text{SincP} \cdot t)$ to $f(a \cdot \text{SincP} \cdot t)$. The curve that dies out in shorter time has the higher SincP value to be specific.

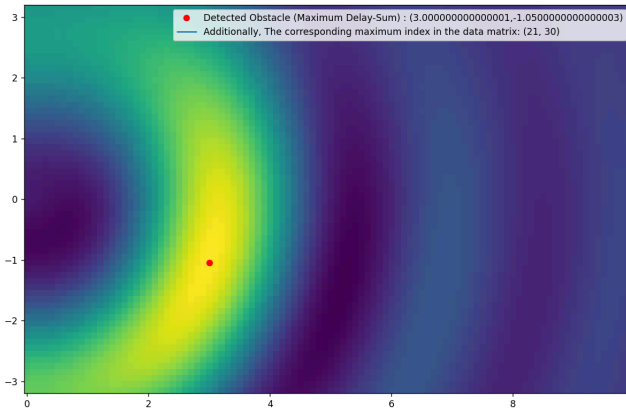


Smaller SincP value

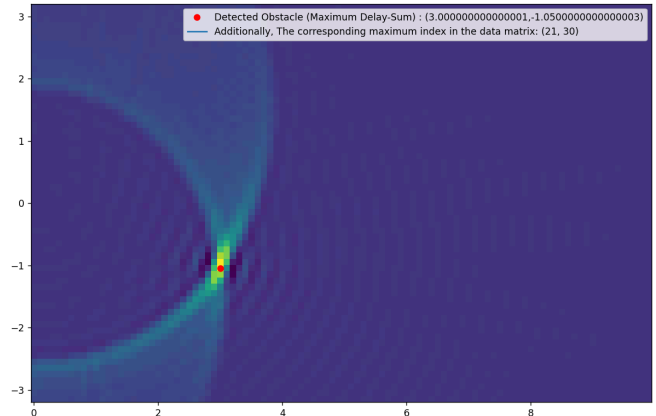


Higher SincP value

- **On the final image, a higher value of SincP from the source will create a sharper estimation of where the obstacle is.** This is because The delay-sum works better if the correct obstacle position gives a steep maxima for the delay-sum, with the surrounding points giving much lower values. And if the signal dies out in a shorter time (Higher SincP), then this is what happens, because we can see that the peak stretches out only over a very short instant in time, which means only the points closest to actual obstacle position will give high values of delay-sum, which is what we want.



SincP = 1



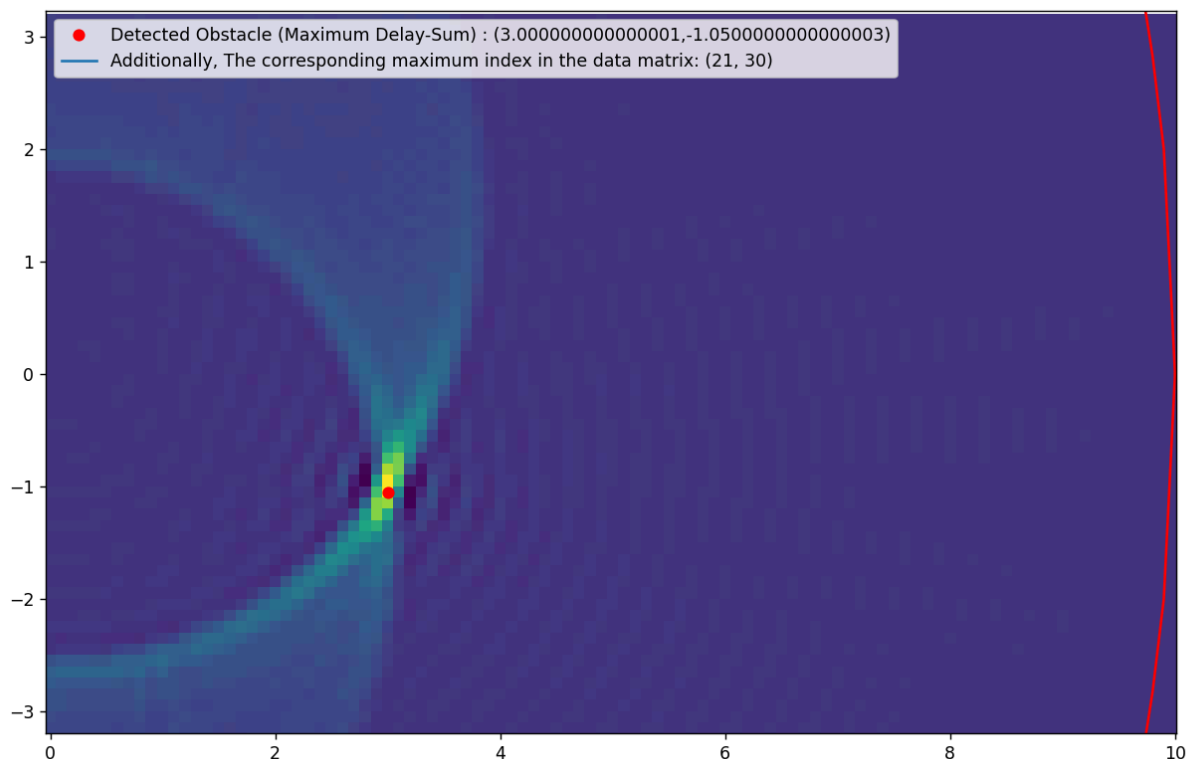
SincP = 10

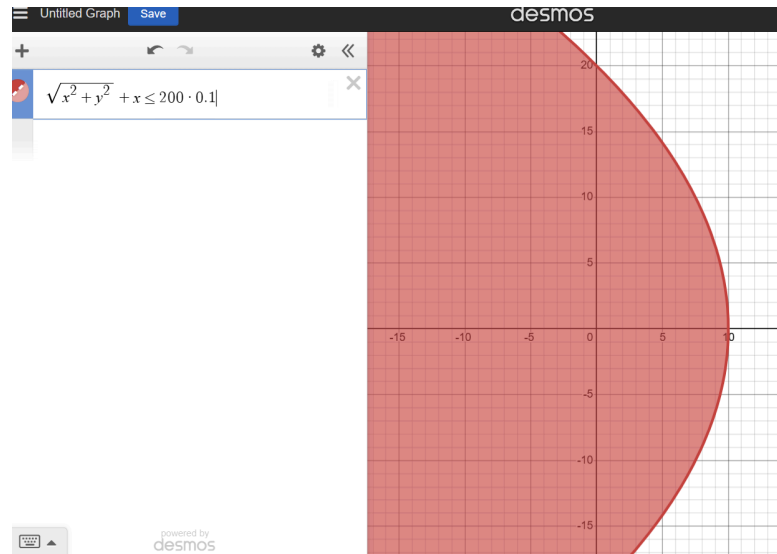
- Q2. Does it make sense to reconstruct(X-Axis) up to Nsamp (For finding the obstacle using DAS)? What value is more reasonable as an upper limit for the x-axis here?
 - A2) A value close to $N_{\text{samp}}/2$ is really the maximum that we need to construct X-Axis till. At $N_{\text{samp}}/2$, the round trip distance from source $(0, 0)$ to $((N_{\text{samp}}/2) * \text{dist_per_samp}, 0)$ to closest mic at $(0, \text{pitch})$ will be close to $2 * ((N_{\text{samp}}/2) * \text{dist_per_samp})$ [assuming pitch is small] =

$N_{\text{samp}} * \text{dist_per_samp}$. This distance of $d = N_{\text{samp}} * \text{dist_per_samp}$ corresponds to **sample number** $\text{int}(d/\text{dist_per_samp}) = N_{\text{samp}}$. Which is the highest sample present in the dataset input given to us for finding the obstacle. (As input data will have N_{mics} rows and N_{samp}).

- This means that $((N_{\text{samp}}/2) * \text{dist_per_samp}, 0)$ is the farthest away obstacle for which any of the mics (as $(0, \text{pitch})$ is the closest mic and it barely captures it, then we say remaining farther away mics won't capture it) captures the $t = 0$ maxima of the source Sinc pulse.
- **At least one of the mics has to capture the $t = 0$ maxima of source Sinc pulse for the Delay sum algorithm to locate the maxima in the given range of points. Hence this X-Limit of $N_{\text{samp}}/2$ is what we need to give.**
- We can actually do a more accurate calculation if accounting for pitch (if pitch is not as small relatively) as well, so the round trip distance from source $(0, 0)$ to $(\text{Max_sample_num} * \text{dist_per_samp}, 0)$ to closest mic at $(0, \text{pitch})$ will be exactly equal to $(\text{Max_sample_num} * \text{dist_per_samp}) + \sqrt{(\text{Max_sample_num} * \text{dist_per_samp})^2 + (\text{pitch})^2}$, and by equating this distance to $N_{\text{samp}} * \text{dist_per_samp}$ (Max allowed distance for DAS), we can get the Max_sample_num which is actually allowed on the X-Axis. But such a calculation was unnecessary is what I felt, since this Max_sample_num will be less than $N_{\text{samp}}/2$ anyway, and not so many samples will be cut by reducing it down to Max_sample_num and so it was simpler to just give $N_{\text{samp}}/2$.
- Q3. The (x, y) coordinates corresponding to the maximum amplitude (yellow color) is approximately $(30, 22)$. Explain why this is the correct expected position for the given obstacle?
 - This is the correct expected position for obstacle $(3, -1)$.
 - $X = 30$ means that it corresponds to 30th sample. So actual X-Coordinate = $30 * \text{dist_per_samp} = 30 * 0.1 = 3$.
 - $Y = 22$ means that it is the 23rd (index starts at 0) mic from the mic at $(-3.15, 0)$ (which is the Mic at the bottom in this arrangement of 64 mics centered at source $(0, 0)$ with pitch 0.1) counting upwards. 23rd mic from bottom = 10th mic from the source counting downwards
 = Y coordinate $(-10 * \text{pitch} + \text{pitch}/2)$ (Pitch/2 added because first mic at $(0, -\text{pitch}/2)$ as it's 64 mics which is even) = $(-1 + 0.05) = (-0.95)$.
 - So the total coordinate = $(3, -0.95)$ which is close to $(3, -1)$. **Y coordinates are discretized to the positions of the mics, that is why it is not exactly (-1) essentially.**

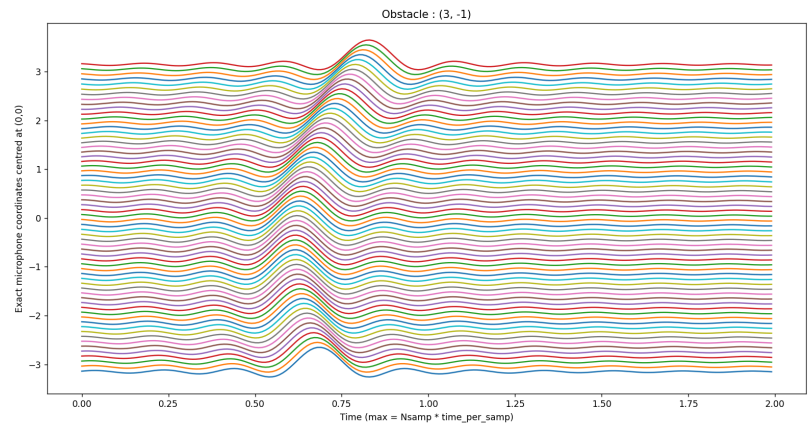
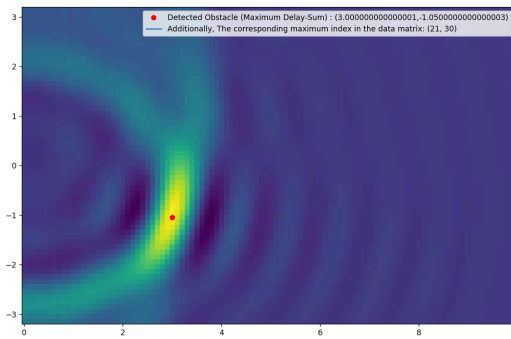
- Q4. What is the maximum obstacle x- and y- coordinate that you can use and still have an image reconstructed?
 - If we assume that the condition for reconstruction is that at least one of the mics detects the t=0 maxima of the Sinc Pulse sent from the source, as before, then:
 - For a general point (x,y) (Talking about point in space like (3,-1), not like (30,22)), the condition(assuming y discretized to pitch(for y coordinate of point = y coordinate of closest mic)) is that:
 $\text{dist}((0,0) \text{ to } (x,y)) + \text{dist}((x,y) \text{ to nearest mic}) < \text{Nsamp} * \text{dist_per_samp}$
 Which is:
 $\text{Sqrt}(x^2 + y^2) + \text{Sqrt}((x-0)^2 + (y-y)^2) < \text{Nsamp} * \text{dist_per_samp},$
- This comes out to be the equation of a parabolic region. Any (x,y) that lies in this parabola will support accurate reconstruction of the obstacle image with the given parameters. Below figure shows said parabola(in red) drawn over our obstacle. = (3,-1) case with all the same parameters as before. (**Note: there is also the restriction of (Lowest_MicY < y < HighestMicY)** along with this because of our choice to discretize y to the mic_positions in this assignment)



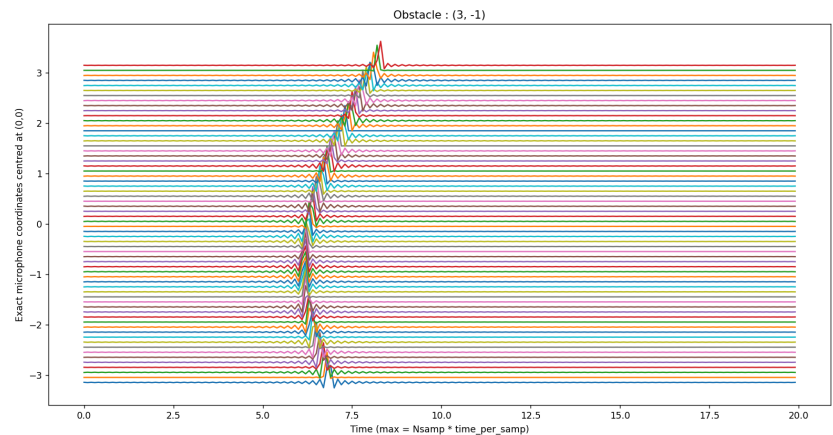
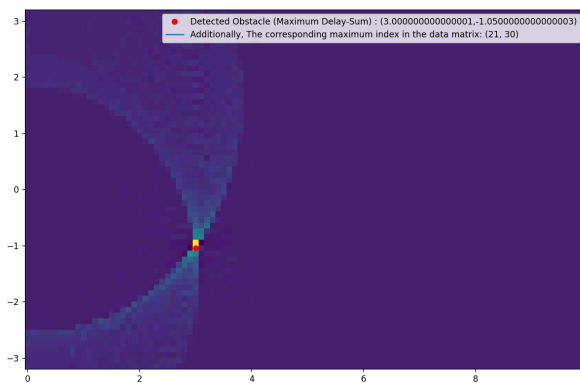


Same locus in desmos.

- Q5) What happens if C is different - if C is decreased it looks like the image becomes sharper. Can you explain why intuitively?

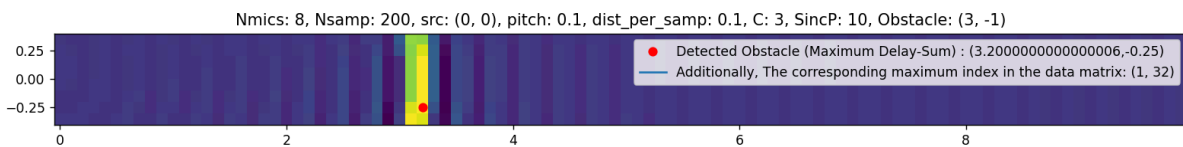
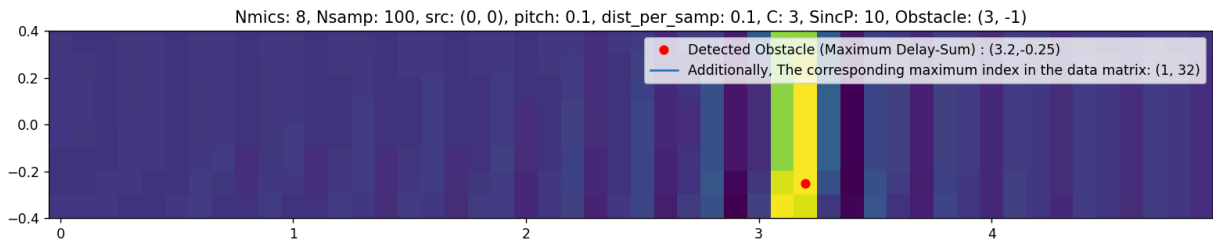
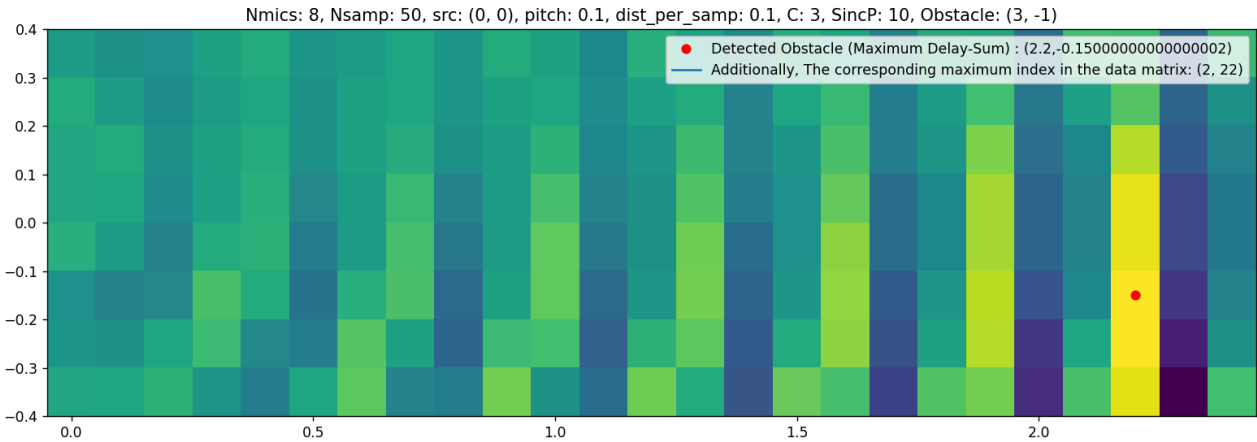


- Above input received by microphone, and final output are taken at $C = 10$ for a set of parameters.

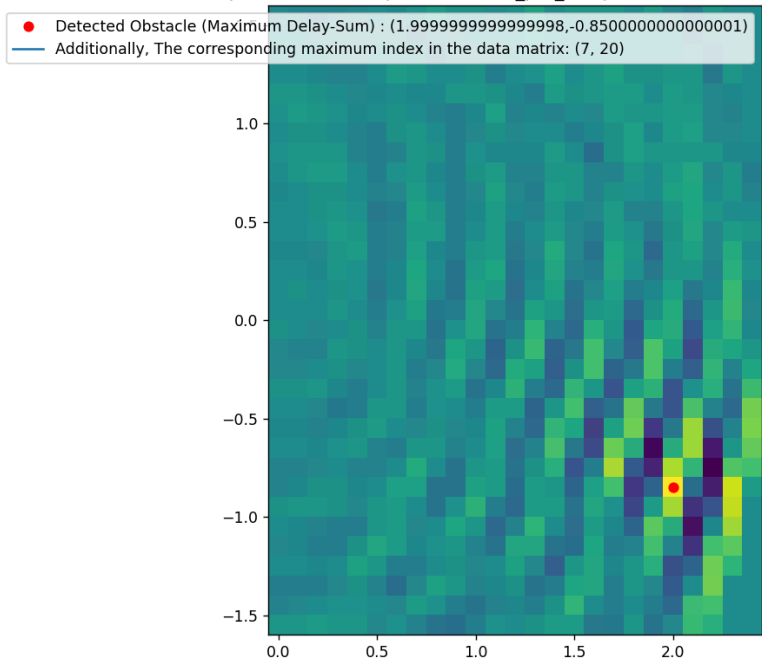


- Above Set is for the same set of parameters, except $C=1$ now.
- We see a more clearly distinguishable final obstacle position in $C = 1$ case. This again has the same reasoning as used in Q1, about the effect of SincP on the image.
- The reason $C = 1$ is sharper is that the input data it receives in the microphones again has very **narrow maxima range in the X-axis and dies out faster**, which means DAS works better as the set of points which gives a high value for delay-sum is pretty small.
- The $C = 10$ input data shows that the input signal at microphones has a pretty wide maxima in the X-Axis. This means that we will get a larger set of points which have high delay-sum, as a large number of points near the actual obstacle line up near the very **wide Maxima** of the input signal when delay-shifted, and so end up getting a pretty high value. That is why the heatmap ends up looking kind of muddled.

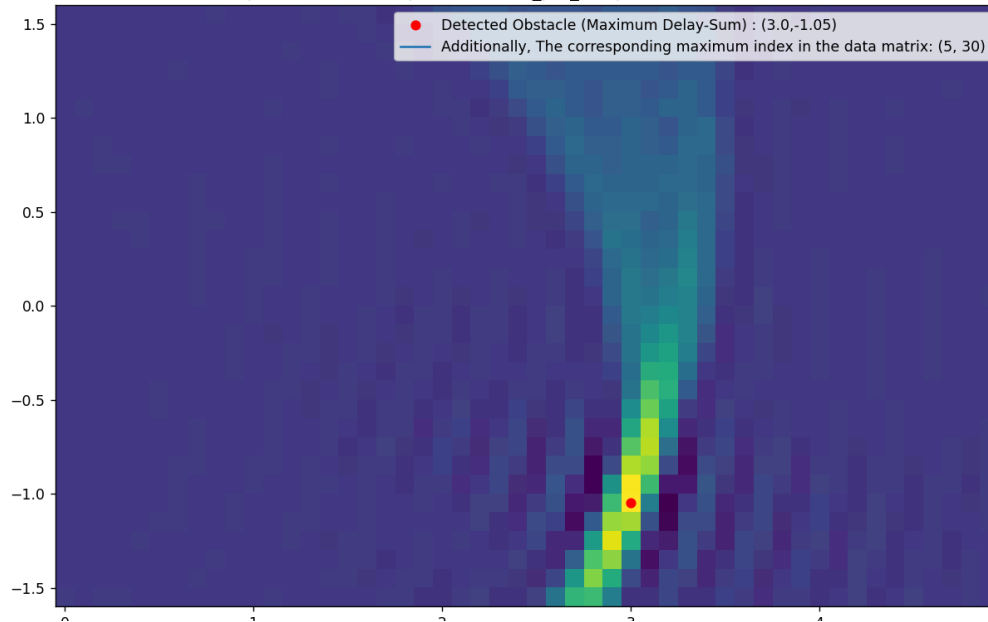
- Q6) What happens if Nmics is increased or decreased? Do the experiments with Nmics = [8, 32, 64] and Nsamp = [50, 100, 200] (all combinations). Attach the resulting images.
 - Increase in Nmics results in increase in the range of Y values we can give for the obstacles and actually reconstruct the obstacle position. Decrease in Nmics creates a corresponding decrease in Y values range.
 - Increase in Nsamp results in increase in the range of X values we can give for the obstacles and actually reconstruct the obstacle position. Decrease in Nsamp creates a corresponding decrease in X values range.
 - Images for each 9 permutations attached below. (Actual obstacle position and other parameters mentioned at top of picture).



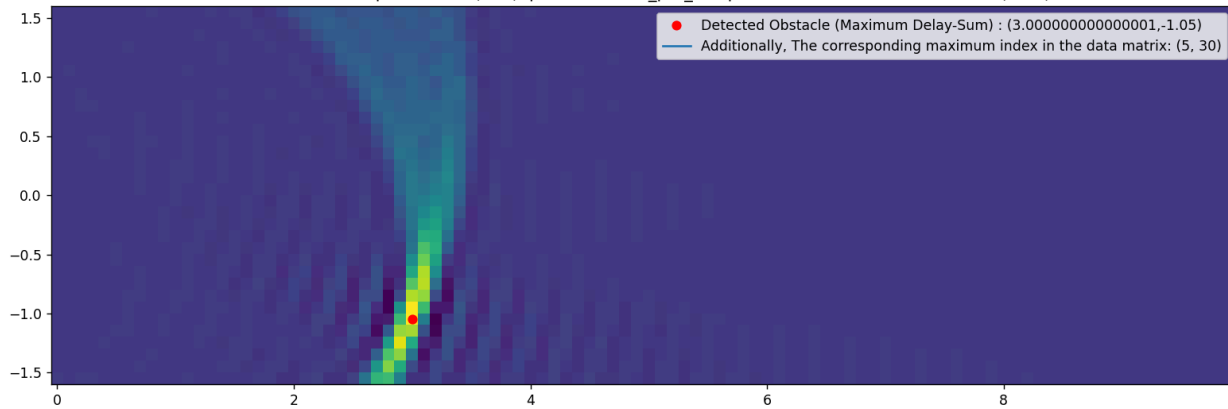
Nmics: 32, Nsamp: 50, src: (0, 0), pitch: 0.1, dist_per_samp: 0.1, C: 3, SincP: 10, Obstacle: (3, -1)



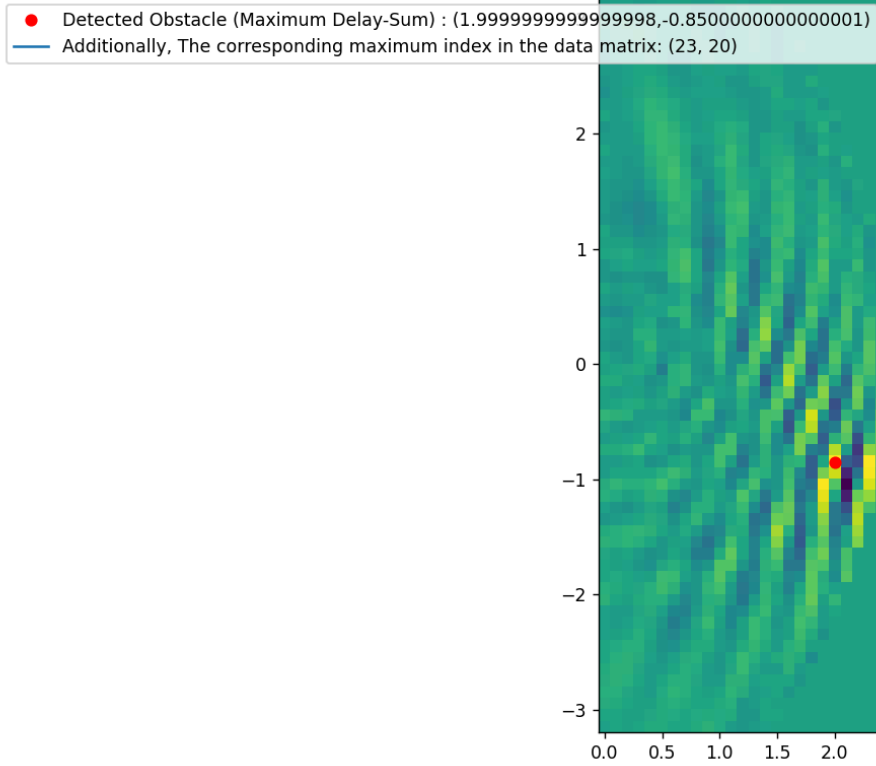
Nmics: 32, Nsamp: 100, src: (0, 0), pitch: 0.1, dist_per_samp: 0.1, C: 3, SincP: 10, Obstacle: (3, -1)



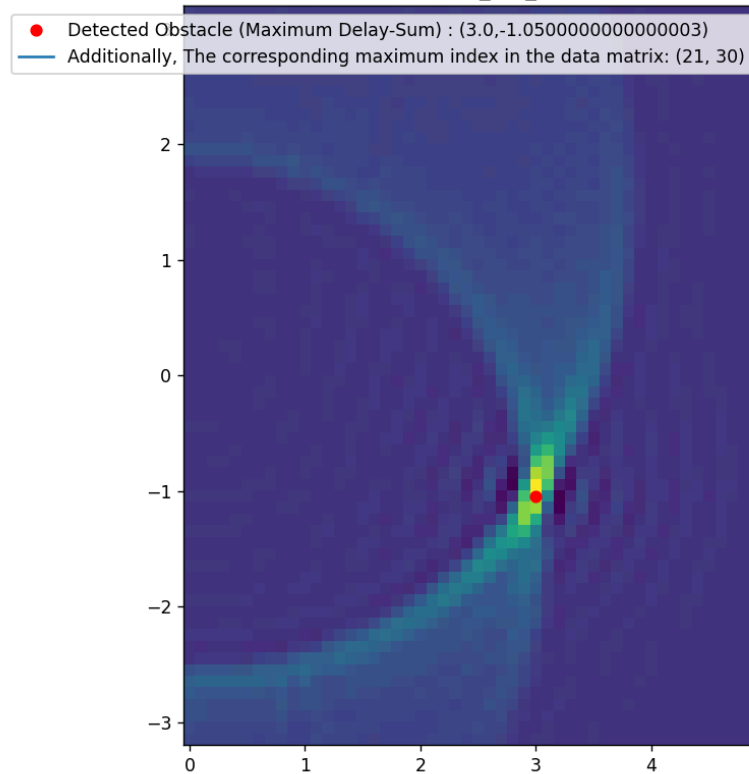
Nmics: 32, Nsamp: 200, src: (0, 0), pitch: 0.1, dist_per_samp: 0.1, C: 3, SincP: 10, Obstacle: (3, -1)

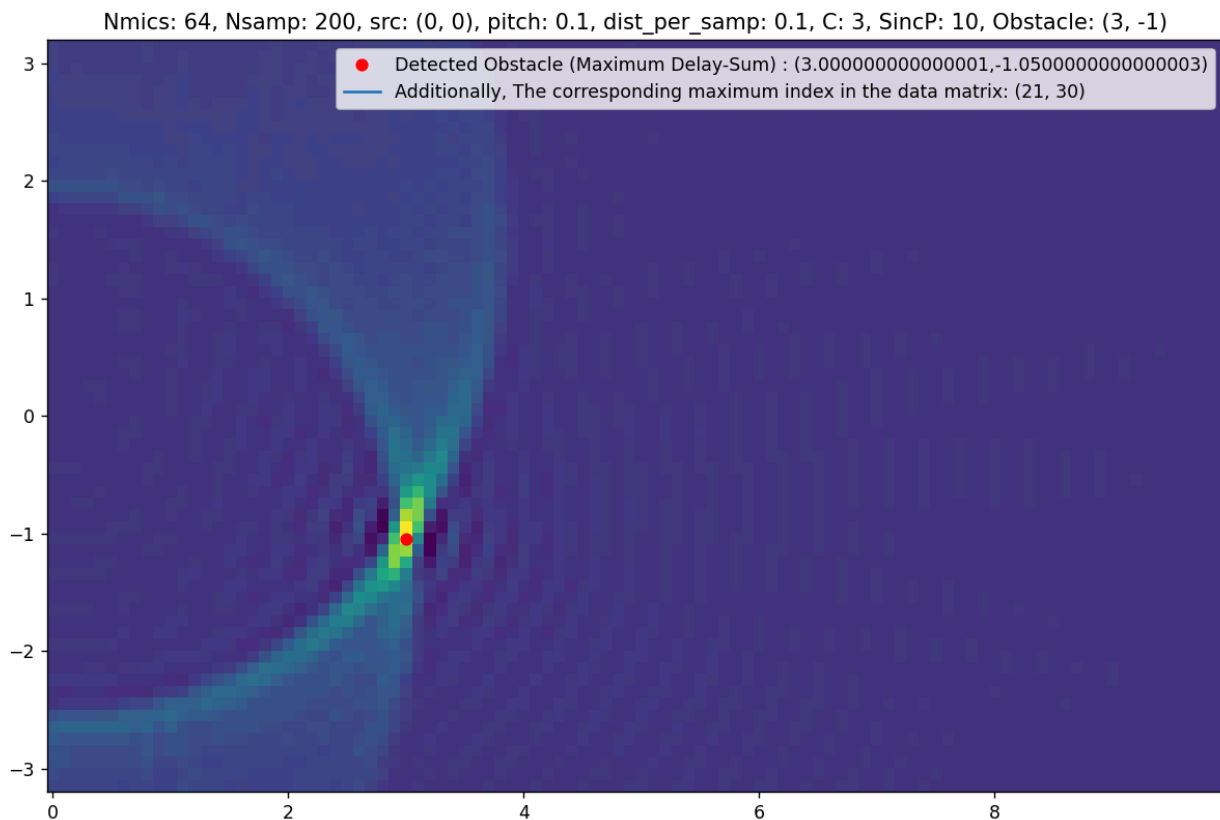


Nmics: 64, Nsamp: 50, src: (0, 0), pitch: 0.1, dist_per_samp: 0.1, C: 3, SincP: 10, Obstacle: (3, -1)



Nmics: 64, Nsamp: 100, src: (0, 0), pitch: 0.1, dist_per_samp: 0.1, C: 3, SincP: 10, Obstacle: (3, -1)





CODE Sections have been done in the Source Code file itself.

Also, all the plots given in the assignment instructions have inverted Y-Axes. I have used the normal Y-Axis instead for all plots.

Hope you read the report, took a long while to make 😭😭